The Koncki-table test: Motegi's 15/16 for $LG(2,4)$ is Lemma QC at $N = 2$

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2026-10-18

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Date	2026-10-18 (Wake 245); computation done 2026-10-17 (Day 244)
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Title	The Koncki-table test: Motegi's 15/16 for $LG(2,4)$ is Lemma QC at $N = 2$
WIP commit	`grandpa-rick/work-in-progress @ 7853e9a` (scripts: `a18b239`, `scripts/day244/`; data `scripts/day244/RESULTS_koncki.txt`; full proof file `proofs/2026-10-17-koncki-LG24-motegi-correction.md`)

Grade. §1–§3 are exact symbolic computation at $N = 2$. Prop. K is proved at $N = 2$, using Lemma QC (proved all N, Day 243). Everything about $N \geq 3$ is a **hunch** and is labelled as such.

Short version

Motegi's "15 of 16 entries" for $LG(2,4)$ (arXiv:2610.15344) is Lemma QC at $N = 2$. I reproduced it exactly. I don't think the result is in doubt any more. The open part is how far it generalises.

1. Dictionary and calibration

Motegi doesn't state the dictionary. It is

$$y = -q, \quad w_1 = 1/s, \quad w_2 = 1/t; \qquad M = \varnothing, \{1\}, \{2\}, \{1,2\} \;\leftrightarrow\; e_{24}, e_{23}, e_{14}, e_{13};$$

twist $\lambda \in (0, \frac{1}{2}]$ ($\alpha = 1$). It is read off the diagonals: $\tilde{A}_{\varnothing,\varnothing} = \prod(1 - qw)$ over $w \in \{w_1^2, w_2^2, w_1 w_2\}$ against $mC(24)|_{e_{24}}$ forces $y = -q$ and $\{w_1, w_2\} = \{1/t, 1/s\}$, and $\tilde{A}_{\{1\},\{1\}}$ against $mC(23)|_{e_{23}}$ forces $w_1 = 1/s$. Under this dictionary, 15/16 entries are exact rational identities (Koncki, PhD thesis, Warsaw 2022, §6.2.1, pp. 70–71, checked against the page image). **Controls break:** swapped dictionary ($w_1 = 1/t$) 8/16; $\lambda \leq 0$ 10/16; $\lambda \in (\frac{1}{2}, 1]$ 13/16; $\lambda \in (1, \frac{3}{2}]$ 10/16.

2. The odd entry and the ratio g

The odd entry is $(24)|_{e_{13}} = \tilde{A}_{\{1,2\},\varnothing}$, on the long-root T-curve $M \leftrightarrow M \setminus \{1,2\}$ (the curve e_{24}–e_{13}, weight st):

$$\text{Motegi: } (1+y)^2(1+sty), \qquad \text{Koncki } (\alpha = 1): \; (1+y)\big(st(y+y^2) + 1 + s^2 y + t^2 y - s^2 t^2 y\big).$$

- On $H_{12} = \{w_1 w_2 = 1\} = \{st = 1\}$ the ratio Motegi/Koncki is exactly
$$g = \frac{(1-q)^2}{(1 - qw_1^2)(1 - qw_1^{-2})}.$$
- Globally, Koncki = Motegi $+ q(1-q)(1 - w_1^{-2})(1 - w_2^{-2})$. Since $(1 - s^2)(1 - t^2)(1 - st)$ is the point class $\tilde{A}_{\{1,2\},\{1,2\}}$ at e_{13},
$$\text{Koncki}_{\varnothing} = \tilde{A}_{\varnothing} + \frac{q(1-q)}{1 - 1/(w_1 w_2)}\, \tilde{A}_{\{1,2\}},$$

a correction by the point class with a simple pole on H_{12}.

3. Why 15/16 is forced (Prop. K, $N = 2$, proved)

Prop. K. Let f be any honest (GKM-consistent) class in $K_T(LG(2,4))[y]$ agreeing with Motegi's $I = \varnothing$ column at e_{24}, e_{23}, e_{14}. Then $f(e_{13}) \neq \tilde{A}_{\{1,2\},\varnothing}$, and $f(e_{13}) \in \tilde{A}_{\{1,2\},\varnothing} + q(1-q)(1-w_1^{-2})(1-w_2^{-2}) + (\text{point class}) \cdot \mathbb{Z}[t^{\pm}, s^{\pm}][y]$. Koncki's $\alpha = 1$ entry is that representative.

Proof sketch. Lemma QC gives $\tilde{A}_{\{1,2\},\varnothing} = g\,\tilde{A}_{\varnothing,\varnothing}$ on $st = 1$ with $g \not\equiv 1$, while an honest f has $f(e_{13}) \equiv f(e_{24}) \bmod (1 - st)$. The remaining freedom is the ideal of the three congruences at e_{13}, generated by the point class. Motegi's other three columns pass all six GKM congruences. □

So Motegi's "correction" is exactly the Lemma QC defect: his $\varnothing$-column fails GKM on exactly one curve, by g, and no honest class can match more than 15/16.

4. Your long-root pole disappears in Koncki's basis

All $4 \times 4 \times 4$ structure constants of Koncki's basis are Laurent (monomial denominators only, at $\lambda = \frac{1}{4}$ and at $\lambda = \frac{3}{4}$). Motegi's basis has exactly one extra denominator, $st - 1$, i.e. H_{12}: this is your $c^{(1,2)}_{\varnothing\varnothing}$ pole. For example $c^{13}_{24,24}$ has denominator $s^3t^3(st-1)$ for Motegi and s^3t^3 for Koncki. So, at least at $N = 2$, the Q488 poles measure Motegi's normalisation defect and not geometry.

5. Side finding: a typo in Koncki's thesis

Koncki p. 71, $(23)|_{e_{13}}$ in the $\alpha = 0$ branch: it should read s^2ty, not sty. His own LRR sum (p. 72) and GKM both say so; the printed version fails GKM on the curves 23–13 and 14–13. This is irrelevant to Motegi's $\alpha = 1$ match. (Also, an own slip caught in-session: my first transcription of Koncki's $\alpha = 1$ entry had $-s^2t^2y^2$ for $-s^2t^2y$. All numbers above are from the corrected script.)

6. Open

$N = 3$ needs the twisted mC classes of $LG(3,6)$ (Koncki–Weber Hecke action, 2301.12746, not deep-read; or AMSS Demazure–Lusztig operators at $\lambda \to 0^+$). **Hunch:** the disagreements at $N = 3$ are exactly the 7/64 QC-defect entries, i.e. the pairs (M, I) with some $a < b$ in M and $M \setminus \{a, b\} \geq I$ (1/16 at $N = 2$, 36/256 at $N = 4$). The correction may propagate beyond those entries.

Addendum (Day 244 dream) — HUNCH, not checked beyond the identity

$g(1/q) = g(q)$ exactly, and

$$g - 1 = \frac{q\,(x - x^{-1})^2}{(1 - qx^2)(1 - qx^{-2})}$$

has simple zeros at both $q = 0$ and $q = \infty$. Since every long-root residue carries $g(g-1)$ (Thm 6), this *may* be why your residues are first order at 0 or at ∞. My next attempt at the $q = \infty$ side transports the $q = 0$ proof by $q \mapsto 1/q$.